Course Title: Programming Language

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Group Members:

Tamanna Rahman

Tasnim Jareen Al Vi

Arnob Barua

Community Issue Tracker

Abstract

The Community Issue Tracker is a console-based application created using the C programming language. Its goal is to improve the reporting and management of community-level issues. In many residential areas, problems like water shortages, power outages, security concerns, or maintenance needs are often communicated informally. This can lead to delays, miscommunication, and the loss of important information. The Community Issue Tracker offers a structured digital solution where residents can register, log in, submit issues, and view past complaints. An administrative module lets authorized personnel monitor users and issues from one central location. The system uses core C programming concepts, including structures, file handling, conditional logic, loops, and menu-driven interfaces. This project shows how basic programming techniques can effectively solve real-world problems.

Introduction

The aim of the Community Issue Tracker project is to design and implement a straightforward yet effective system that helps residents report issues in an organized way. The project focuses on turning a common real-life problem into a software solution using basic programming knowledge.

In many neighborhoods, management relies on manual complaint handling methods like phone calls, written notes, or verbal reports. These methods are often inefficient and lack proper tracking. Complaints may be forgotten, duplicated, or ignored without a central system. This causes frustration for residents and makes it hard for management to identify ongoing or urgent problems.

The Community Issue Tracker tackles these challenges by providing a platform where every issue is recorded and stored permanently. Each complaint is connected to a registered user, improving accountability and transparency. From an academic standpoint, this project is important because it applies theoretical concepts from programming courses to a practical situation. It helps students learn how software systems are planned, built, tested, and refined.

System Overview

The Community Issue Tracker is designed as a menu-driven console application. The system operates through a series of clear menus that guide users through various operations. There are two main roles in the system: community users and administrators.

Community users are regular residents who can register, log in, and submit issues. Administrators have higher-level access that lets them view all registered users and reported issues. The system utilizes file handling to store data, ensuring that information remains available even after the program closes. This simulates essential database behavior using text files, making it suitable for those with beginner to intermediate programming skills.

Features and Implementation

User Registration System

The user registration system allows new residents to create an account in the Community Issue Tracker. During registration, users must provide crucial information like their name, flat or house number, and a password. This information uniquely identifies each user and ties them to the issues they report.

From an implementation perspective, this feature is developed using a C structure to logically group all the user related information into a single unit.

typedef struct

{ char

username[20]; char

flat[10]; char

password[20];

} User;

File handling functions, such as fopen, fprintf, and fclose, are used to store user information in a file. Each time a new user registers, their data is added to the file, ensuring existing records are not overwritten. This feature introduces persistent storage and shows how data can be saved beyond program execution.

User Login and Authentication

The login system guarantees that only registered users can access the core features of the application. Users must enter their username, flat number, and password correctly to log in. This helps maintain privacy and prevents unauthorized access.

The login process reads stored user data from the registration file and compares it with the credentials entered by the user. String comparison functions validate input. If the credentials match an existing record, the user gains access to the main menu. If not, an error message appears. This feature illustrates conditional logic, loops, and basic authentication principles.

Issue Reporting Module

The issue reporting module is the main functionality of the Community Issue Tracker. Once logged in, users can submit complaints about various categories, including water supply, electricity, security, and general maintenance. Each issue includes details provided by the user, helping administrators understand the problem's nature.

This module uses a menu-based selection system that lets users choose the type of issue they wish to report. The issue details, along with the user's identification information, are saved in a file using structured records. A dedicated C structure is used to store all the issue-related data into a single record.

```
typedef struct
{    int ID;    char
type[20];    char
description[200];    char
status[50];    char
reportedBy[30];
} Issue;
```

File handling ensures that every reported issue is stored permanently. This feature shows how structured data can be recorded and managed effectively.

Issue Viewing System

The issue viewing system allows users to check the issues they have previously reported. This feature enhances transparency and gives users confidence that their complaints have been documented. It also helps prevent users from submitting duplicate complaints for the same problem.

Implementation involves reading issue records from the file and displaying them in an easy-to-understand format. This feature reinforces file reading techniques and formatted output. It also emphasizes the importance of organizing stored data, making it easy to retrieve and present clearly.

Admin Panel

The admin panel provides administrative control over the system. Administrators can view all registered users and all reported issues across the community. This feature is crucial for monitoring recurring problems and managing resources effectively.

The admin panel is secured through a separate authentication mechanism. Only authorized credentials can access administrative functions. This part of the system uses conditional checks and restricted menu options. It demonstrates role-based access control and system-level management.

Moderator Panel

The Moderator Panel is a key part of the Community Issue Tracker. It allows controlled access between regular users and full administrators. The moderator role helps monitor reported issues without granting complete administrative rights. Moderators review submitted issues to ensure complaints are valid, properly sorted, and free from irrelevant or duplicate entries. This helps maintain data quality and prevents misuse of the system. Moderators can also spot recurring problems in the community and flag them for administrative attention. Implementation Details The moderator panel uses role-based access control within the system. After logging in, the program checks the user’s role and

shows menu options specific to moderators. Conditional statements restrict access, allowing only authorized moderators to enter this panel. Moderators can view all reported issues saved in the issue file through file reading operations. They cannot change user credentials or delete records. This protects the system's integrity and prevents unauthorized changes. This separation of duties boosts security and mirrors real-world software design practices. The Moderator Panel shows how to use structured decisionmaking, file handling, and access control logic in C programming. It also emphasizes how different user roles can be managed within a single application.

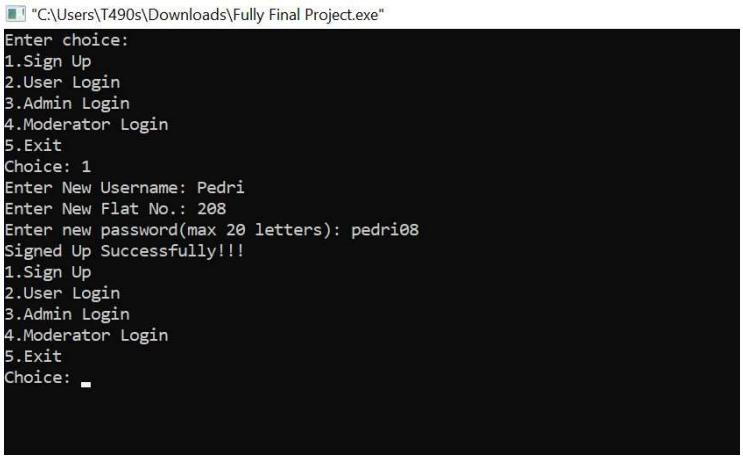
Menu Driven Interface

The entire Community Issue Tracker runs through a menu-driven interface. Menus guide users through registration, login, issue reporting, and viewing options step by step. This method makes the system user-friendly, even for those with limited technical knowledge.

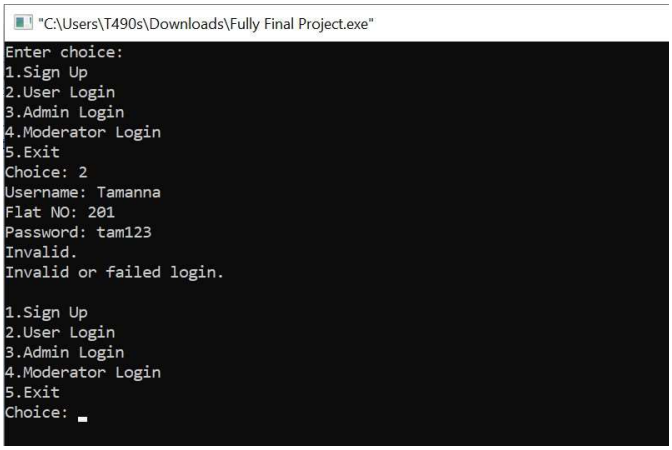
From an implementation perspective, switch-case statements and loops control program flow. The menu system ensures smooth navigation and prevents unexpected program termination. This feature connects all other components of the system.

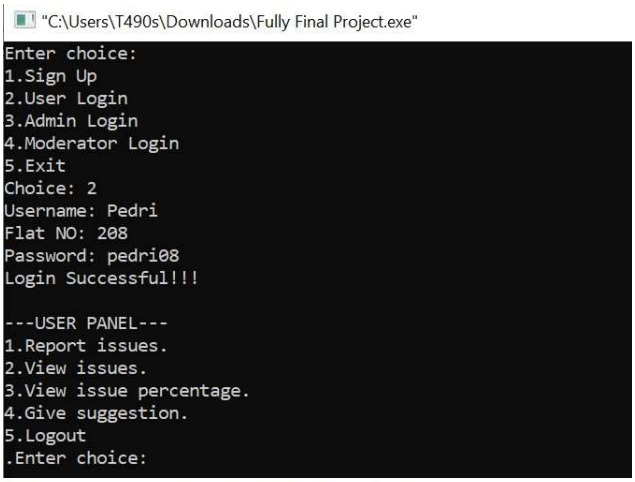
Testing and Results

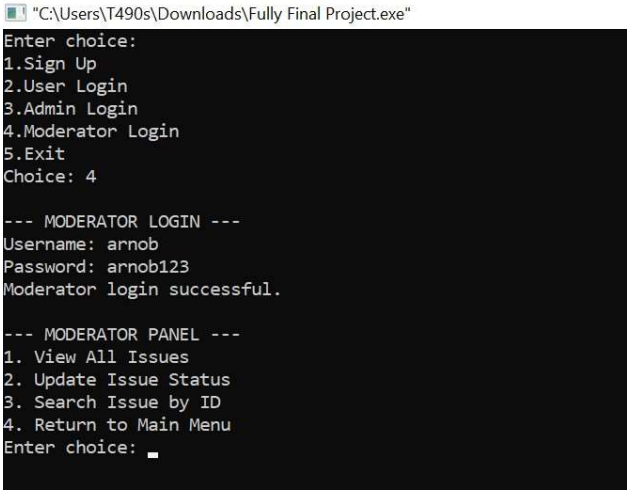
The Community Issue Tracker underwent multiple tests to verify its functionality. User registration was tested with valid and invalid inputs.



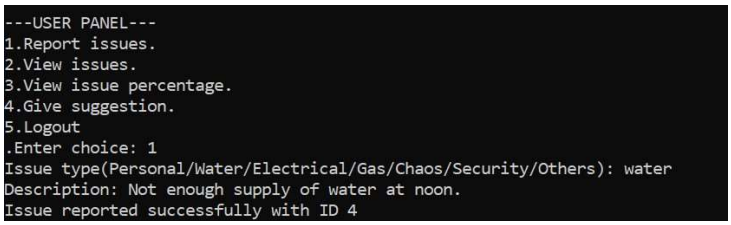
Login authentication was tested with correct and incorrect credentials.

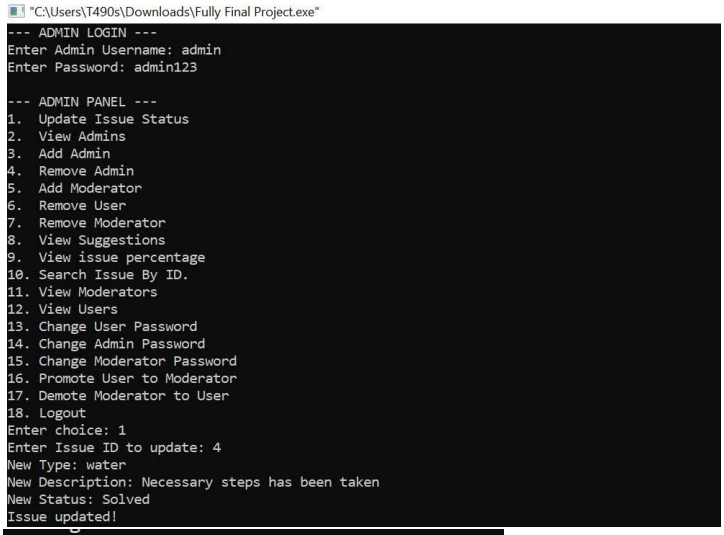


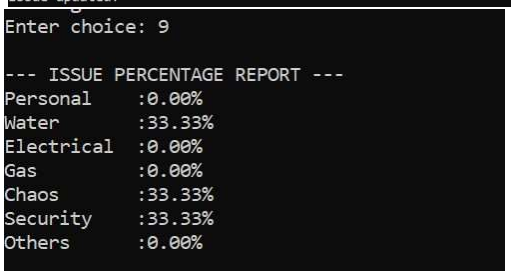




Issue reporting was tested across different categories, and file storage was checked to confirm data persistence.







The results indicated that the system worked as expected. User data and issue records were stored correctly and retrieved without errors. The admin panel and moderator panel successfully displayed all records. Minor input handling issues were identified and corrected during testing. Overall, the system proved stable and reliable for its intended purpose.

Challenges and Solutions

Initially issues occurred in opening and reading files correctly. Then input line buffer was another big challenging part (specially in login part).Using fscanf and fgets together caused the major buffer issue. File handling issue was tackled by using proper file formats and

files were closed after each operation. Input buffer was cleared and newline characters were handled properly.

Another challenge was managing user input, particularly handling strings and avoiding buffer-related errors. This was tackled by improving input validation and carefully managing input functions. Designing smooth menu navigation also needed multiple revisions to avoid infinite loops or accidental exits.

Conclusion

The Community Issue Tracker project successfully shows how a real-world community issue can be addressed using basic C programming techniques. Through this project, students gained practical experience in structures, file handling, authentication systems, and menu-driven program design. The project emphasizes the importance of planning, testing, and continuous improvement in software development.

Future improvements could include password protection, issue status tracking, priority levels for urgent complaints, solved issues removed, and . Overall, this project offered valuable insight into how programming can be used to create meaningful and practical applications.